Prism Gate Optical Ring Photon Reader

Concept: Levi Burd · Engineering design document · Oct 27, 2026

The design in one page

The Prism Gate Optical Ring Photon Reader is a near-self-imaging optical storage ring whose diagonal fold mirrors are the reader stations. Light enters through the six-inch frame, is switched into a 5.6 m ring by a Pockels-cell gate, circulates past the stations once every 18.7 ns, and is switched out on command.

The ring is buildable from known parts. Free-space "loop-and-switch" memories already store polarization qubits coherently (Pittman and Franson 2002), though early versions lost about 19% per cycle. The lowest-loss loop of this type reached 1.2% per cycle, but stored only one known polarization (Kaneda and Kwiat 2019). The Prism Gate's contribution is to combine the two, qubit-preserving switching at near-1% loss, and to make the ring's fold mirrors into measurement sites.

Parameter	Design value
Wavelength	780 nm (matches rubidium-based nondestructive detectors and published loop memories)
Circuit length and time	5.6 m, 18.7 ns
Ring optics	Two concave mirrors, R = 3.0 m, 2.8 m apart along each arm; six flat diagonal fold mirrors
Beam radius at the curved mirrors	0.84 mm
Switch	Pockels cell inside a polarizing Sagnac, so any polarization qubit is switched coherently
Loss target	1.4% per circuit with conservative parts, 1.0% target
Stations	Six fold sites for low-loss leakage monitors; a quantum herald only if one exists that accepts nanosecond photons

Two numbers decide the quantum track, and an independent referee review (Section 8) found that the first one blocks today's technology:

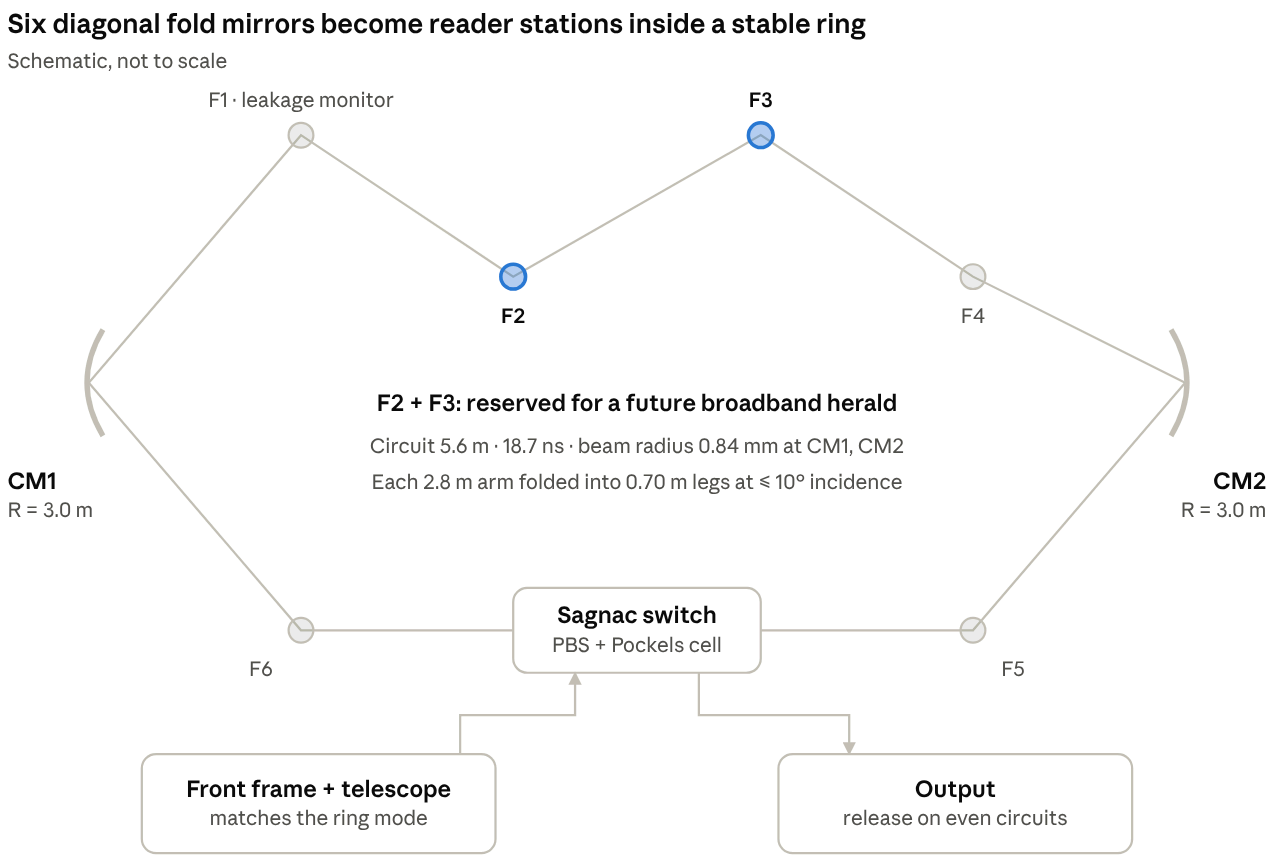
1. **Bandwidth.** A photon must be shorter than the 18.7 ns circuit, so any herald inside the ring must accept photons of roughly 0.1 GHz bandwidth or more. The demonstrated atom–cavity heralds work at a few MHz, with photons hundreds of nanoseconds or more long, far longer than the ring. No herald demonstrated so far fits inside it.
2. **Survival.** Even with a fitting station, two heralds beat one only if each station's photon survival exceeds $1/(2 - \eta_h)$, where η_h is herald efficiency given survival. With the best published $\eta_h = 0.87$, that bar is 0.885; published survival is 0.66 without qubit preservation and 0.31 with it.

So the design is staged accordingly: the ring as a qubit-preserving, self-monitoring storage loop is buildable now, and it defines the exact specification a future herald must meet.

Every number in this document can be tested live in the companion Prism Gate Ring Designer: move the arm length, mirror reflectance or station figures and it recomputes stability, beam size, loss per circuit and the cascade verdict.

1. Why this geometry

A storage ring is the only arrangement where light passes the same angled walls many times by design, so it is the natural home for the original idea of readers built into diagonal baffles.



Prism Gate ring layout · two curved mirrors, six fold stations, one switch

Two concave mirrors refocus the beam on every circuit, so it never spreads out of the stations' acceptance; the flat diagonal folds only bend the path, so each can host a station without disturbing the focusing. The Sagnac switch admits light from the front frame and releases it to the output.

Two numbers follow from this geometry. Because the curved mirrors refocus the beam on every circuit, hold length is limited by loss, not diffraction. And the 18.7 ns circuit leaves room for a Pockels cell to switch between passes.

2. Optical design numbers

The ring's round-trip phase is set 8° short of perfect imaging on purpose: an exact 4f relay sits on the edge of optical stability, so small tilts can push it marginally unstable. All values below come from ABCD ray-transfer calculations at 780 nm.

2.1 Why not exact self-imaging

Two concave mirrors of focal length f joined by two arms of length 2f form a 4f relay: after each circuit every ray returns inverted, and after two circuits it returns to itself. The round-trip stability parameter is exactly (A + D)/2 = −1, the boundary of stability. Tilting the curved mirrors by 3° in crossed planes pushes it to −1.000004. The practical effect is small: beam size grows only about 0.3% per circuit, about 12% over 40 circuits. Perfect imaging is also the most forgiving geometry for an input beam that does not match the ring's own mode. The reason to back off is margin against larger misalignment, not a crisis at 3°.

Shortening each arm from 3.0 m to 2.8 m moves the ring inside the stable region with margin:

Arm length	Circuit	(A + D)/2	Round-trip Gouy phase	Beam radius at curved mirrors
3.0 m	6.0 m, 20.0 ns	−1.000	180° (edge of stability)	undefined
2.9 m	5.8 m, 19.4 ns	−0.998	176°	0.85 mm
2.8 m (chosen)	**5.6 m, 18.7 ns**	**−0.991**	**172°**	**0.84 mm**
2.7 m	5.4 m, 18.0 ns	−0.980	169°	0.82 mm

At 2.8 m the ring stays formally stable over a wide range of curved-mirror tilt, but large tilts distort it: at 20° in crossed planes the beam at the curved mirrors shrinks from 0.83 mm to about 0.40 mm, which ruins mode matching. Astigmatism at the design's 3° is negligible, with effective focal lengths of 1.498 m and 1.502 m. The ring still nearly images, which keeps mode mismatch from building up over many circuits.

2.2 Geometry

1. Curved mirrors: R = 3.0 m (f = 1.5 m), used at 3° incidence or less to keep astigmatism negligible.
2. Each 2.8 m arm is folded into four 0.70 m legs by three flat mirrors, giving six flat diagonal baffles in all and an enclosure about 0.75 m × 0.30 m.

3. Fold mirrors use coatings specified for low s–p retardance at their angle, measured on arrival. Shallow incidence does not by itself guarantee low retardance: some dielectric coatings show several degrees even near normal incidence. A calibrated compensating waveplate removes what remains. Folding in orthogonal planes cancels retardance only for matched coatings, and it makes the ring non-planar, which adds an image and polarization rotation that must be modelled.
4. Every clear aperture keeps a radius of at least 2.5 times the local beam radius: about 2.1 mm at 0.84 mm, which clips under 10^{-5} per aperture. The Pockels crystal is the tightest aperture (often 3–4 mm across), so its clear aperture is checked first.
5. Glass counts. The Pockels crystal and beamsplitter cube shorten the effective focusing distance by $L(1 - 1/n)$ and lengthen the delay by $L(n - 1)$, and the Sagnac switch adds its own path. The final arm length is set on the effective distance, and the real circuit time is somewhat longer than 18.7 ns.

2.3 Beam into the ring

The six-inch front frame and collection lens must deliver a beam matched to the ring's own mode: a waist and position taken from the 2.8 m design, set with a two-lens telescope and checked by beam profiling at two planes. Mismatch here is paid once, not every circuit, which is why the collection optics are allowed a larger loss budget than the ring itself.

3. The switch

The gate is a Pockels cell inside a polarizing Sagnac interferometer, the arrangement Pittman and Franson used to switch polarization qubits in and out of a free-space loop without destroying their coherence. This removes the obvious objection to a polarization-based switch: that it would read the very polarization the experiment is trying to preserve.

3.1 How it works

A plain polarizing beamsplitter plus Pockels cell routes photons by polarization, so it can only store a known polarization; this is the version used for heralded single-photon sources (Pittman, Jacobs and Franson 2002). In the Sagnac version, the beamsplitter splits the qubit into its horizontal and vertical parts, which travel the same loop in opposite directions and both pass the Pockels cell. Switching acts on both parts alike, so

the stored state is not measured. The same group has since stored one photon of a polarization-entangled pair in this kind of memory and tested Bell's inequality before and after storage (Evans et al. 2023).

Two cautions come with this precedent. First, the stored qubit is bit-flipped on alternate circuits (Pittman and Franson 2002), so the control system tracks circuit-count parity and releases on even circuits or applies a matching correction. Second, these early loops were lossy, roughly 19% to 26% per cycle; they prove the switching concept, not low loss. The Sagnac beamsplitter and Pockels cell sit inside the ring, so their loss and polarization errors are paid on every circuit.

3.2 Timing budget

The Pockels cell must finish switching within one circuit:

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\tau_{rise} + \tau_{photon} + \tau_{jitter} < \tau_{circuit} = 18.7\
\text{ns}
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Photons from spontaneous parametric down-conversion last picoseconds, so the photon term is negligible. The binding term is the cell's rise time. A rise time of roughly 10 ns was measured in an early loop of this kind (Pittman, Jacobs and Franson 2002), leaving about 8 ns for trigger jitter and cable delay. The ring was sized to 18.7 ns for exactly this reason.

3.3 Switch specification

Property	Requirement	Why
Rise and fall time	10 ns or less, 10–90%	Fits the 18.7 ns circuit with margin
Repetition	Two switch events per stored photon, at the source repetition rate	One event to capture, one to release
Insertion loss	0.5% or less per pass, AR-coated at 780 nm	Dominates the per-circuit budget after the mirrors
Extinction while on	Set before Stage A from the target fidelity	Residual leakage lowers stored-qubit fidelity every circuit

Property	Requirement	Why
Trigger	From the herald detector or the pump-laser clock, never from the stored photon	The gate must not sample the signal

Cell crystal, driver and exact figures come from vendor datasheets during procurement; these rows are the acceptance criteria those datasheets must meet.

4. Loss budget

With conservative parts the ring loses about 1.4% per circuit; with target parts, about 1.0%. The closest measured comparison is Kaneda and Kwiat's 1.2% per cycle: 0.8% from the Pockels cell and 0.2% from each of two concave mirrors, with no folds and no Sagnac beamsplitter (Kaneda and Kwiat 2019). That loop stored one known polarization, so 1.2% is a floor for a simpler device, not a demonstrated figure for qubit storage. This ring adds six folds and a Sagnac switch, so matching it is a real engineering target.

4.1 Per circuit

Element	Count per circuit	Conservative transmission	Target transmission
Flat diagonal fold mirrors	6	0.9995 each	0.9998 each
Concave mirrors, R = 3.0 m	2	0.9995 each	0.9998 each
Pockels cell	1	0.995	0.996
Sagnac beamsplitter passes	1	0.995	0.996
Circuit total		**0.986 (1.4% loss)**	**0.990 (1.0% loss)**

The switch, not the mirrors, dominates. Any deliberate tap adds directly to this per-circuit loss: a 1% tap at one fold would nearly double the budget and cut 40-circuit survival from 0.57 to 0.38. That is why the monitor stations in Section 5.1 read the natural leakage of high-reflectivity mirrors instead.

4.2 One-time terms

Element	Assumed transmission	Note
Collection and mode matching into the ring	0.90	Paid once; set by the front-frame telescope
Capture and release switching	0.98	Two switch events per photon
Output coupling into fiber or free-space receiver	0.95	Measured against a no-device baseline
One-time total	**0.84**	

4.3 What a hold costs

Survival over a hold, before the one-time terms:

Circuits held	Hold time	At 1.4% (conservative)	At 1.0% (target)
10	187 ns	0.868	0.904
20	374 ns	0.754	0.818
40	748 ns	0.570	0.669

Stage A's pass criterion is a measured per-circuit loss of 1.5% or less.

4.4 Polarization fidelity per circuit

Loss is only half the budget. Every circuit also adds a small polarization error from three sources: Pockels-cell voltage error and ringing, the beamsplitter's limited extinction (typically about 1:1000 in transmission and 1:100 in reflection), and leftover mirror retardance. Random errors add roughly linearly with circuits, so a 40-circuit hold at about 96% fidelity needs about 10^{-3} infidelity per circuit or better. Systematic errors add in amplitude, growing about N-fold, so they must be calibrated out with a compensating waveplate rather than averaged away. This budget is measured in Stage C by process tomography after 2, 10 and 40 circuits.

5. Stations

The classical station is a sensor behind a diagonal baffle, reading the light that leaks through it. The quantum station is a herald at a fold, and the ring sets two hard requirements any such herald must meet.

5.1 Classical station: the ring reads its own loss through mirror leakage

Every high-reflectivity fold mirror leaks a little light, typically 10 to 100 parts per million. An amplified photodiode behind one fold records that leakage each time a bright pulse passes. The pulse train decays by the ring's per-circuit transmission, so the station measures ring loss directly, circuit by circuit, while adding almost no loss itself. With bright pulses the leaked signal is ample. This is the Stage A–B instrument: one ring-down measurement tests both the routing model and the readout. A deliberate 1% tap is kept only as a removable alignment aid.

5.2 Quantum station: a herald at a fold

The best-demonstrated qubit-preserving nondestructive detection of traveling optical photons reflects the photon off a cavity holding a single atom in a superposition, then reads the phase the photon left on the atom (Reiserer, Ritter and Rempe 2013; Niemietz et al. 2021). Two separate atoms have also detected the same photon twice (Distante et al. 2021).

The bandwidth blocker. These stations work with photons a few MHz wide, which last hundreds of nanoseconds or more. The ring's circuit is 18.7 ns and the switch needs about 10 ns of it, so a stored photon must last a few nanoseconds: roughly 0.1 GHz of bandwidth or more. A MHz photon is tens of meters long or more and simply does not fit in a 5.6 m ring. Narrowing a broadband photon to MHz would destroy it for this purpose, and a ring long enough for MHz photons would be hundreds of meters, with prohibitive loss.

So no demonstrated herald can sit inside this ring. The ring instead defines the specification for one: qubit-preserving, nondestructive, accepting photons of 0.1 GHz or more, and returning them to the ring mode with survival above the Section 5.3 bar. This document did not find a demonstrated device that meets it. Until one exists, the cascade question is tested with stations in series along a path, as Distante et al. did, not inside this ring.

5.3 When two stations beat one

For stations with herald efficiency η_h (given that the photon survives) and photon survival s, the probability that a photon is heralded and still exits is $s \cdot \eta_h$ for one station and $s^2[1 - (1 - \eta_h)^2]$ for two. Two win only when:

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s > \frac{1}{2 - \eta_h}
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Station	η_h given survival	Survival s	Required s	Qubit preserved
Reiserer et al. 2013	0.87	0.66	0.885	No
Niemietz et al. 2021	0.79	0.31	0.83	Yes, 96.2% fidelity
Hypothetical future station	0.90	—	0.91	—

Reiserer et al. report 74% unconditional efficiency, which corresponds to 87% given survival; the formula needs the conditional figure.

No published station clears the bar. At the demonstrated qubit-preserving survival of 0.31, a cascade is pointless, and even one station loses 69% of photons. Survival per station, not herald efficiency, is the number the field must move.

5.4 Why one station cannot simply be reused each circuit

A tempting shortcut is to let the photon pass the same atom–cavity station on every circuit. It fails for two reasons:

1. **The herald phase cancels.** The photon imprints a π phase on the atom; a second pass adds another π, returning the atom to where it started. The atom must be read out and reset between passes.
2. **Reset costs circuits.** Each circuit is 18.7 ns, and atomic fluorescence readout and re-preparation usually take several microseconds. Even an optimistic 1 µs means about 54 circuits of waiting, with survival 0.47 at 1.4% per circuit. A second pass then needs $s > 1.9$, which is impossible.

The photon-number measurement is a parity measurement, so this holds for any station of this type, not only atoms. Multiple heralds, if they ever fit in the ring, must be separate devices at separate folds; F2 and F3 are reserved for them.

6. Parts, cost and schedule

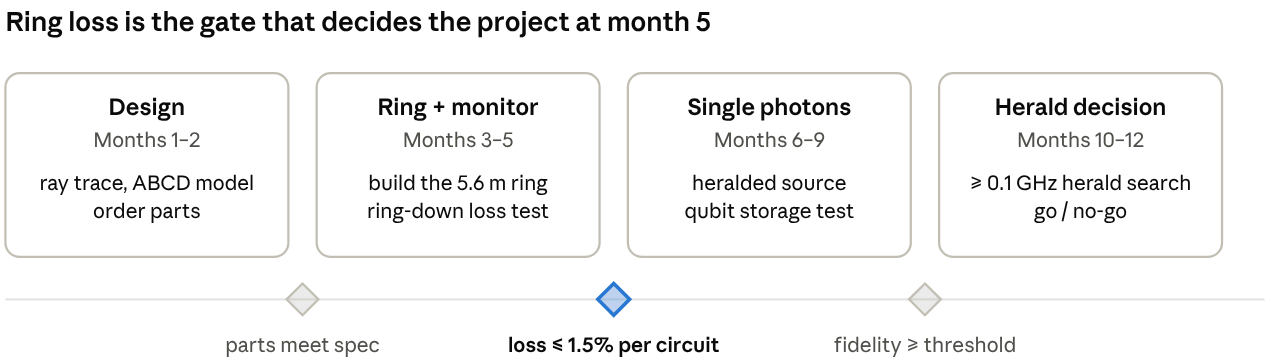
The ring and its monitor stations are a one-year, one-lab project built from catalogue optics. The quantum herald is a specification and a search at month 10, not a build.

6.1 Bill of materials

Item	Qty	Specification	Cost class
Flat fold mirrors, 1 inch	6	Dielectric HR at 780 nm, R ⩾ 0.9995 at ⩽ 10°, low-scatter polish	Low
Concave mirrors, R = 3.0 m, 1 inch	2	Same coating, used at ⩽ 3°	Low
Leakage monitor	1	Amplified photodiode behind an HR fold with measured leakage of 10–100 ppm; removable 1% tap for alignment only	Low
Pockels cell and high-voltage driver	1	Section 3.3 table	High (largest single item)
Polarizing beamsplitter and waveplates	1 set	Sagnac switch, AR at 780 nm	Low
Kinematic mounts, posts, breadboard, enclosure	1 set	Light-tight, about 0.8 m × 0.4 m footprint	Medium
780 nm laser and fast photodiode	1 each	Pulsed or modulated for ring-down	Medium
Beam profiler	1	For mode matching and the 2.5w aperture check	Medium
Heralded photon-pair source at 780 nm	1	Down-conversion, or borrowed from a partner lab	High
Single-photon detectors and time tagger	2–4 channels	Stage C onward	High

Cost classes are relative and deliberately unpriced; replace them with vendor quotes in month 1. A lab that already owns detectors and a pair source pays mainly for the Pockels cell and the optics.

6.2 Schedule



12-month schedule · four phases, three gates

The month-5 gate decides the project: if the ring cannot reach 1.5% loss per circuit, no station placed in it can be useful, and the work stops with a clean, publishable loss measurement.

7. What still has to be true

Five results decide whether the Prism Gate is real; each has a stop rule fixed now, before any data exist.

Must be true	Measured at	Stop rule
Switch fits the circuit, including glass path	Month 3, fast photodiode on the gated beam	Lengthen the arms once (each extra 0.3 m per arm adds 2 ns); stop if still too slow
Ring loss near the 1.2% floor	Month 5, leakage-monitor ring-down	Stop if loss exceeds 1.5% per circuit after two full realignments
Stored qubits keep their state	Month 9, process tomography after 2, 10 and 40 circuits	Stop the quantum track if per-circuit infidelity exceeds about 10^{-3} after compensation

Must be true	Measured at	Stop rule
A herald exists that accepts photons of 0.1 GHz or more	Month 10, literature and partner search	If none, publish the ring as a qubit memory and the herald specification; do not claim in-ring heralding
Such a herald clears $s > 1/(2 - \eta_h)$	After a fitting herald exists	Report the single-station result; do not claim a cascade advantage

Three honest limits remain whatever the results:

1. No demonstrated herald fits a nanosecond ring; the in-ring quantum station waits on a broadband nondestructive detector that does not yet exist.
2. The device cannot read an unknown qubit without disturbing it; it heralds presence and stores state.
3. Its delay and loss are measurable by anyone on the link, so it is never invisible.

Why this is the strongest honest version

Every part is now either demonstrated in the literature or reduced to one measured number with a stop rule, and the one thing that does not yet exist is named and specified rather than assumed. Its central idea, readers built into angled walls that light passes repeatedly, is realized as leakage-monitor stations in a stable, qubit-preserving storage ring. That ring is useful on its own and is ready for a broadband herald the day one is demonstrated.

8. Independent review

An independent referee pass checked every number in this design with ABCD calculations and against the cited papers. The optics were confirmed; the findings below are built into the design.

Review finding	How the design handles it
Atom–cavity heralds need MHz photons, far longer than the 18.7 ns ring	The herald is a specification, not a build; cascades are tested outside the ring

Review finding	How the design handles it
Reiserer's 74% efficiency is unconditional; 87% given survival	Cascade bar set at 0.885 using the conditional figure
The 1.2% loss result came from a loop storing one known polarization	Treated as a floor for a simpler device, not a qubit-memory result
Survival must be quoted at this ring's own loss	Survival table given at 1.4% and 1.0% per circuit
A deliberate 1% tap would nearly double ring loss	Monitors read HR-mirror leakage of 10–100 ppm
Polarization errors accumulate every circuit	10^{-3} per-circuit fidelity budget with a stop rule
Exact imaging is only marginally unstable; large tilts distort the mode	Both quantified in Section 2
Glass in the crystal and cube changes focusing and delay	Included in the geometry rules
Atomic readout and reset take several microseconds	Reusing one station on successive passes ruled out

References

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All ring-optics figures (stability parameter, Gouy phase, beam radius, tilt tolerance, survival over circuits) were computed with ABCD matrices for this document and should be re-run in the Stage A ray trace.